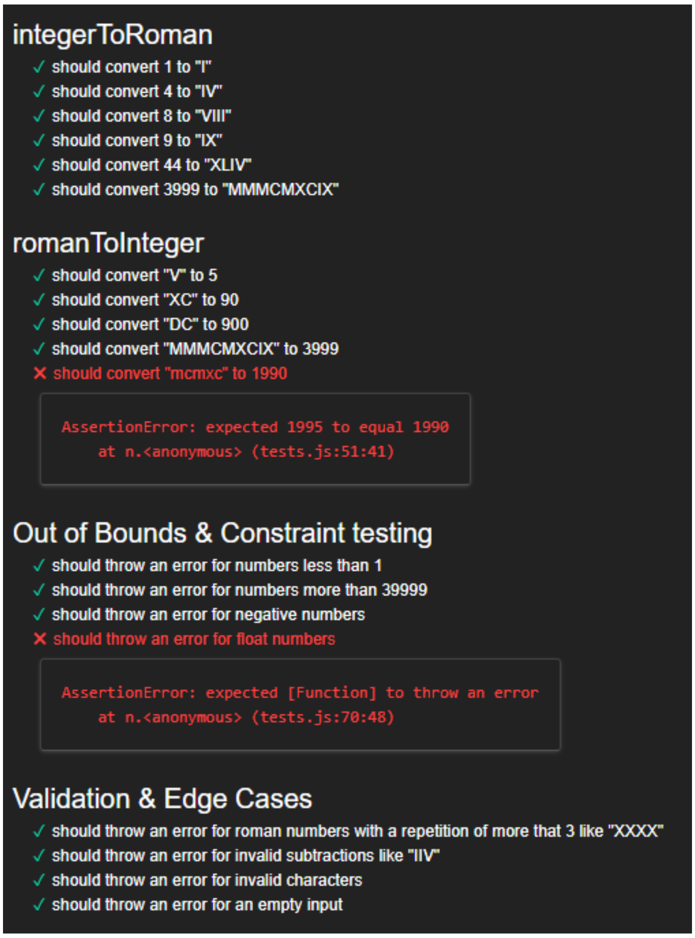
Alvaro Garcia

Task #4 - Roman Converter - Test Automation

For this task I used all the test cases that the AI provided me in the last task, this is a snapshot of the automated test cases:



```
integerToRoman
  ✓ should convert 1 to "I"
  ✓ should convert 4 to "IV"
  ✓ should convert 8 to "VIII"
  ✓ should convert 9 to "IX"
  ✓ should convert 44 to "XLIV"
  ✓ should convert 3999 to "MMMCMXCIX"

romanToInteger
  ✓ should convert "V" to 5
  ✓ should convert "XC" to 90
  ✓ should convert "DC" to 900
  ✓ should convert "MMMCMXCIX" to 3999
  ✗ should convert "mcmxc" to 1990

    AssertionError: expected 1995 to equal 1990
        at n.<anonymous> (tests.js:51:41)

Out of Bounds & Constraint testing
  ✓ should throw an error for numbers less than 1
  ✓ should throw an error for numbers more than 39999
  ✓ should throw an error for negative numbers
  ✗ should throw an error for float numbers

    AssertionError: expected [Function] to throw an error
        at n.<anonymous> (tests.js:70:48)

Validation & Edge Cases
  ✓ should throw an error for roman numbers with a repetition of more that 3 like "XXXX"
  ✓ should throw an error for invalid subtractions like "IIV"
  ✓ should throw an error for invalid characters
  ✓ should throw an error for an empty input
```

As we can see there are two test cases that are marked as incorrect because the actual output is different than the expected output. The first one is that, as the AI in the previous task said, a roman number in lowercase should be accepted and transformed into an integer, and the program is not implemented with this precondition. The second failure is that it does not check if a integer is a float, this is a more important failure as in the preconditions in the previous task was indicated that non integer and fraction numbers should not be represented as roman numerals, in this case, if we input a float like “10.5”, the program returns “X” instead of "Please enter a valid integer number.".

To resolve both failures, we should change the code into in the “romanToInteger” function to first transform every character into uppercase and then transforming it as usual. And for the second failure, before transforming it into a roman numeral it should check if the input number is a float, and if so, print an error: "Please enter a valid integer number.".